

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th November 2020

**Subject CP1 – Actuarial Practice
(Paper A)**

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

- Q. 1)** A general insurance company selling property insurance expects a liability outgo (including expenses) of Rs 900 Million and premium, coupon/redemption inflow of Rs 1000 Million in the next financial year.
- i)** Should the company have liquid assets in its portfolio? If yes, why? (3)
 - ii)** State three key reasons why a general insurance company might hold high levels of liquid assets in its portfolio compared to its immediate expected liability outgo? (3)
- [6]**
- Q. 2)** The regulator has mandated a life insurer selling traditional participating and non-participating products to stop writing future new business after it failed to satisfy solvency criteria laid out by the regulator. Set out the short-term and long –term effects of closure of new business on the insurance company’s
- i)** Withdrawal rates (5)
 - ii)** Investment strategy (5)
- [10]**
- Q. 3)** Due to the increase in hospitalizations from a vector borne disease during monsoon season, CEO of a small health insurer company has asked you, the Product Actuary, whether the company could come up with a simple health product covering risks arising from this disease.
- i)** Propose a suitable product design. (5)
 - ii)** Suggest a suitable reinsurance program for the product. (5)
- [10]**
- Q. 4)** A general insurance company is currently facing capital constraints due to credit defaults in its investment portfolio. Suggest ways in which it could improve its capital position, explaining the advantages and disadvantages of each option. **[12]**
- Q. 5)**
- i)** State the principles of investment for a provider of defined benefit schemes. (2)
 - ii)** Outline key factors that the trustees of a defined benefit need to consider when finalizing a fund’s investment strategy. (4)
- Due to recent financial turbulence in the stock markets some blue chip stocks have fallen by more than 30% in the last 3 months. The investment manager of a defined benefit scheme, which invests in debt and cash, feels that it is the appropriate time to invest in the equity markets as it would give enhanced return on assets in the medium term to long term.
- iii)** Discuss the pros and cons of this suggestion of the investment manager? (8)
- [14]**
- Q. 6)** LVM sells electrical cars and started its operations 5 years back. It had then launched a “Reward Scheme” under which the company would bear cost of all repairs for first six years

from the date of sale of a new vehicle. The Company has observed that the actual cost of repairs claimed by clients have exceeded the budgeted cost for this scheme every year since its inception.

You are working in a consulting actuarial firm, which has recently been appointed by LVM to review the current Scheme.

- i) List data that you would require to perform analysis to understand the reasons for the deviation. (4)
 - ii) You have been asked to prepare a report on improving the scheme design overall and recommending suitable other mitigations to reduce cost to company. Outline the points that you would cover in your report. (12)
- [16]**

Q. 7) A health insurer offers a critical illness insurance product. The demand for the product has been growing in the recent past. The Sales Director has expressed his unhappiness over the Turn Around Time (TAT) for policy issuance and he believes that the principal reason for the delay is the initial underwriting process.

Hence the Sales Director has suggested that, in order to cut costs and increase TAT for issuance, the level of initial underwriting that the Company carries out should be reduced.

- i) Outline the reasons for performing initial underwriting for a critical illness product? (5)
 - ii) Discuss the advantages and disadvantages of this suggestion of Sales Director. (11)
- [16]**

Q. 8) A mid-sized general insurance company uses triangulation technique (basic chain ladder method) for determination of IBNR (Incurred But Not Reported) reserves for its motor liability portfolio.

- i) Outline the checks you would perform on the claims data collected for determination of IBNR reserves before performing the reserving exercise. (5)

The CFO of the company feels that the claims data is getting distorted due to a few large claims in the past few years and impacting IBNR reserves calculated. You have been asked to examine whether triangulation should be done by removing the large claims from the data.

- ii) Critically analyse the CFO's statement. (8)
 - iii) Discuss the key considerations that need to be taken into account in fixing a threshold for bi-furcation of data into large and "normal" claims. (3)
- [16]**
